The Web of Bonds: How the DICTONIC Bonds Hand Over

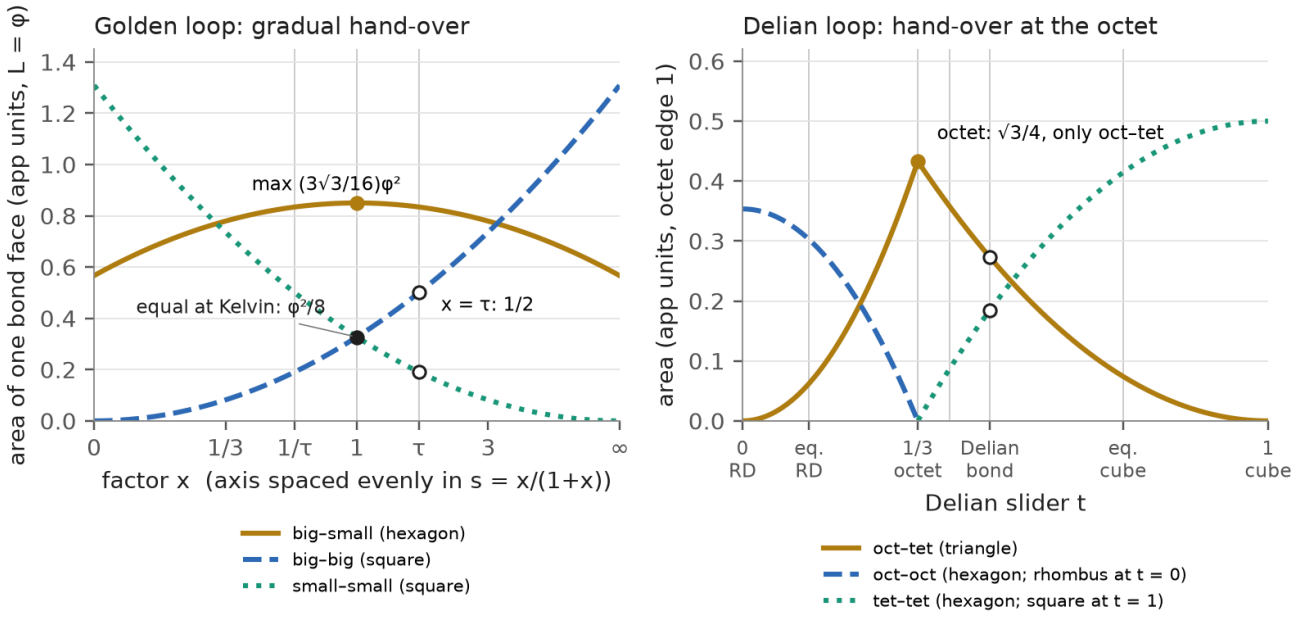


(working title — DICTO decides)

Theorem: both DICTONIC loops are weighted Voronoi (power) diagrams of one fixed crowd of points of two kinds, with one weight slid between the kinds; the bonds are the edges of the dual weighted-Delaunay web; the bond between unlike pieces never breaks inside a loop, while the two like bonds hand over, gradually in the golden loop and all at once, at the octet, in the Delian loop

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Paper #52 (proposed; DICTO decides) · Draft, version 1 · 25 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/7b0fd8b/studies/2026-10-25-bond-relay and …/studies/2026-10-25-physical-anchors (research branch, commit 7b0fd8b)
The ideas are DICTO's: the DICTONIC SOLIDS and the DICTONIC BONDS and their names, and the web of bonds, a network of bonds that "should create a web of lattices that change with the rhythm of the loop" (DICTO, 25 October 2026); computation, exact proofs and drafting with Claude (Anthropic).

In short. DICTO's two loops of space-filling pairs, the golden loop of paper #48 and the Delian loop of paper #51, each slide one pair of solids through a whole family of shapes. Draw one line, a bond, between every two pieces that share a face, and the lines make a web. We show that each loop is really one fixed crowd of points of two kinds, the piece centres, and that the only thing that moves is one number, a weight given to one kind against the other. Every piece is the region nearer (in this weighted sense) to its own point than to any other. Then the web is exactly the web of the classical weighted Delaunay construction. As the weight slides, the bond between unlike pieces never breaks. The two like bonds hand over: in the golden loop gradually, both alive all the way and equal at Kelvin's cell; in the Delian loop all at once, at the octet, where every rhombus of a hidden rhombic dodecahedron trades its long diagonal for its short one. Read physically, the weight is a ball radius squared: the golden loop is the caesium chloride structure, the Delian loop the fluorite structure, and the range where the balls can touch is bounded exactly by Pauling's classical radius-ratio limits.
Main theorem. For every x in [0, ∞] the golden network is the power diagram of the body-centred cubic sites with weight difference $(\varphi^2/4)(x-1)/(x+1)$; for every t in [0, 1] the Delian network is the power diagram of the fluorite sites with weight difference $(3/8)(2t-1)$. The web is the 1-skeleton of the dual regular subdivision, with exact bond areas. In the golden loop all three bond types live on the whole open loop, with constant face lattice; the two like bonds have area ratio x^2 and are equal only at $x = 1$. In the Delian loop the unlike bond lives exactly on (0, 1), oct–oct exactly on [0, 1/3), tet–tet exactly on (1/3, 1]; $t = 1/3$, the octet, is the only parameter at which both pieces exist and only one bond type is present. **Theorem 2 (the balls):** the weight is a squared radius; with unlike neighbours touching, $\rho = (x+2)/(2x+1)$ and $\rho = t/(1-t)$, and the touching windows end exactly at Pauling's limits $\sqrt{3}-1$ and $\sqrt{(3/2)}-1$. All results are exact (371 + 61 checks, none failed).



DICTO

Figure 1. The exact area of one bond face along each loop (Main theorem; Table 1). Left, golden loop (axis evenly spaced in s = x/(1 + x), so the whole loop 0 ≤ x ≤ ∞ fits): the unlike bond (gold) never breaks and peaks at Kelvin; the two like bonds trade size, equal at Kelvin (φ²/8) and 1/2 against 1/(2φ²) at x = τ. Right, Delian loop: oct–oct (blue) dies exactly where tet–tet (green) is born, at the octet t = 1/3, where the unlike bond peaks, with a corner, at the octet triangle √3/4. Open circles: x = τ and the Delian bond.

1. Where it started

> *DICTO's DICTONIC world in Kaleidohedra shows a web: one line for every face two pieces share. DICTO's idea was that this web should itself be a lattice that changes with the rhythm of the loop. We asked what that web is, exactly.*

Paper #48 [1] cut the gaps between icosahedra at the golden point and placed the resulting pair, the Icosaxes and the Unborn Axes, in a one-parameter family T(x), T(1/x) of truncated octahedra (Koca's orbit [10]). Paper #51 [2] cut the octet by one sliding plane per tetrahedron–octahedron pair and found the Delian bond, Samson and Delilah. DICTO called the solids the DICTONIC SOLIDS and the pairs the DICTONIC BONDS, and in the app each family runs as a loop. On 25 October 2026 DICTO proposed that the network of bonds should create a web of lattices that change with the rhythm of the loop. The app draws that web (Figure 5): one line per shared face, thicker for a larger face. Float runs on a small network suggested that bonds trade places along each loop. Two exact studies [S1, S2] answer what the web is, and this paper reports them.

2. The two loops

> *Both loops are pairs of pieces that fill space. Each piece sits on a point; the points never move. Only the shapes change.*

Golden loop. T(x) is the convex hull of all permutations of (±(1 + x), ±x, 0), scaled so that its squares lie at distance L/2 from its centre, with L = φ (paper #48, Theorem 4). T(x) sits on the big sites B = φℤ³, T(1/x) on the small sites S = φℤ³ + (φ/2)(1, 1, 1), for x from 0 to ∞. We write s = x/(1 + x) in [0, 1]: then T(x) is the set of signed permutations of (1, s, 0)·L/2, and T(1/x) the same with 1 − s in place of s. At x = 1 both are Kelvin's truncated octahedron [20]; at x = τ they are the Icosaxes and the Unborn Axes.

Delian loop. The pieces of paper #51 at slider t in [0, 1], with a = 3(1 − t)/2: the tetrahedron pieces sit on Q = (ℤ + ½)³/√2, the octahedron pieces on O = {n ∈ ℤ³ : $n_1 + n_2 + n_3$ odd}/√2, octet edge 1. At t = 0 the octahedron pieces are Kepler's rhombic dodecahedra, at t = 1/3 the octet, at t = 1 the tetrahedron pieces are cubes; in between lie the two equal-edged members, the half-and-half member and the Delian bond (Samson and Delilah).

Power diagram. Given sites c_i with weights w_i, the power cell of c_i is the set of points p with $|p - c_i|^2 - w_i \le |p - c_j|^2 - w_j$ for all j: a Voronoi cell in which each site is given a head start [3, 4]. Two cells meet in a plane perpendicular to the line between their sites (the power bisector). The cells are convex and fill space. Their dual, the weighted (regular) Delaunay subdivision, has an edge between two sites exactly when their cells share a face [3].

3. Main theorem

> *Each loop is one fixed crowd of points with one weight slid between the two kinds. The web of bonds is the classical dual web. The unlike bond never breaks; the like bonds hand over.*

> **Main theorem.** (i) For every x ∈ [0, ∞] the golden network is the power diagram of the body-centred cubic sites B ∪ S with $w_B - w_S = (\varphi^2/4)(x - 1)/(x + 1) = (L/2)^2(2s - 1)$. For every t ∈ [0, 1] the Delian network is the power diagram of Q ∪ O with $w_Q - w_O = (3/8)(2t - 1)$. At t = 0 the Q cells are single points, at t = 1 the O cells are single points.
> (ii) The web of bonds is the 1-skeleton of the dual regular subdivision, and the bond faces and their areas are those of Table 1.
> (iii) *Golden: gradual hand-over.* For 0 < x < ∞ all three bond types are present and the face lattice is constant. Big–big rises strictly from 0 and small–small falls strictly to 0; their ratio is x^2, the diagonals of the two square faces (sL and (1 − s)L) add up to the lattice spacing φ, and the two areas are equal only at x = 1. The whole picture is symmetric under x ↔ 1/x. *Delian: hand-over at the octet.* Oct–tet is present exactly on (0, 1), oct–oct exactly on [0, 1/3), tet–tet exactly on (1/3, 1]. t = 1/3 is the only parameter at which both pieces exist and every face is of one bond type (there the web is bipartite). The ends t = 0 and t = 1 carry one bond type only because one piece has shrunk to a point. The golden loop never has a single bond type: at x = 0 there are two (big–small and small–small), at x = ∞ two (big–small and big–big).

Loop	Bond	Face	Area (app units)	Alive	Shape of the curve
golden (L = φ)	big–small	hexagon (triangle at x = 0, ∞)	$(\sqrt{3}/8)\varphi^2 \cdot (x^2 + 4x + 1)/(1 + x)^2$	whole loop	rises on (0, 1), falls on (1, ∞); max $(3\sqrt{3}/16)\varphi^2 \approx 0.8502$ at Kelvin; $(\sqrt{3}/8)\varphi^2 \approx 0.5668$ at both ends
	big–big	square, diagonal sL	$\varphi^2 x^2/(2(1 + x)^2)$	born at $x = 0^+$	strictly increasing, $\to \varphi^2/2$ as $x \to \infty$
	small–small	square, diagonal (1 − s)L	$\varphi^2/(2(1 + x)^2)$	dies as $x \to \infty$	strictly decreasing from $\varphi^2/2$ at x = 0
Delian (octet edge 1)	oct–tet	triangle	$(9\sqrt{3}/4)t^2$ for $t \le 1/3$; $(9\sqrt{3}/16)(1 - t)^2$ for $t \ge 1/3$	exactly (0, 1)	rises to √3/4 (the octet triangle) at 1/3, then falls; a corner at 1/3, slopes 3√3/2 and −3√3/4
	oct–oct	hexagon (rhombus at t = 0)	$(\sqrt{2}/4)(1 - 9t^2)$	exactly [0, 1/3)	strictly decreasing from √2/4 (the rhombic dodecahedron's rhombus)
	tet–tet	hexagon (square at t = 1)	$(3t - 1)(5 - 3t)/8 = (1 - a^2)/2$	exactly (1/3, 1]	strictly increasing to 1/2 (the cube's face)

Table 1. The bond faces and their exact areas. Sample values: at Kelvin (x = 1) big–big = small–small = $\varphi^2/8 \approx 0.32725$; at x = τ big–big = 1/2, small–small = $1/(2\varphi^2) \approx 0.19098$, big–small = $(\sqrt{3} + 3\sqrt{15})/16 \approx 0.8344$; at the Delian bond oct–tet = $2^{1/3}\sqrt{3}/8 \approx 0.2728$ and tet–tet = $1/2 - 2^{1/3}/4 \approx 0.1850$.

> *Why: choose the weights so that every flat face of every piece is exactly the plane of equal weighted distance between the two points it separates. Then each piece is its point's weighted cell, and the rest is the classical duality.*

Proof. (i) With the weights in (i), every facet plane of every piece of the app is the power bisector of the two sites that share it, for every s ∈ (0, 1) and every t in each of the regimes (0, 1/3) and (1/3, 1), as an identity in the parameter [S1, A2]. Every corner of the piece satisfies every other power inequality against every site within a fixed distance, with slack ≥ 0 on the whole open regime: the slacks are linear in s or t and have no root inside [S1, A1]. The facet planes' own vertex enumeration gives back exactly the piece's corners, and their normals sum to 0 with rank 3, so they bound a bounded region; together with A1 this is piece = power cell. A uniform distance bound shows that sites farther away cannot cut the cell [S1, A7]. The regime boundaries and the ends x = 0, ∞, t = 0, 1/3, 1 are checked by building the power cells from scratch among the fixed sites, with no use of the app's pieces, in exact arithmetic, and comparing with the app's pieces; at t = 0 and t = 1 the vanishing kind's cell is a single point [S1, B1]. (ii) Power diagrams are dual to regular subdivisions [3], so two sites are bonded exactly when they span an edge of a dual cell. The face of each bond is computed in closed form [S1, A3] and checked at every sample, rational and irrational [S1, B2], and against the app's own web to 10^{-9} [S1, C2]. (iii) The monotonicity, birth and death points and the corner at 1/3 follow from the closed forms [S1, A6]; the ratio of the like bonds is $s^2/(1-s)^2 = x^2$, so they are equal only at x = 1; the square diagonals are sL and (1 − s)L, which add to L = φ. The swap: s ↔ 1 − s flips the sign of the weight, and the translation by (φ/2)(1, 1, 1) exchanges B and S, so the diagram at 1/x is the diagram at x translated with big and small exchanged; hence big–big(1/x) = small–small(x) and big–small is unchanged. The bond types present in each regime are read off the incidences of the dual cells (Corollary 3) [S1, B6]. ∎

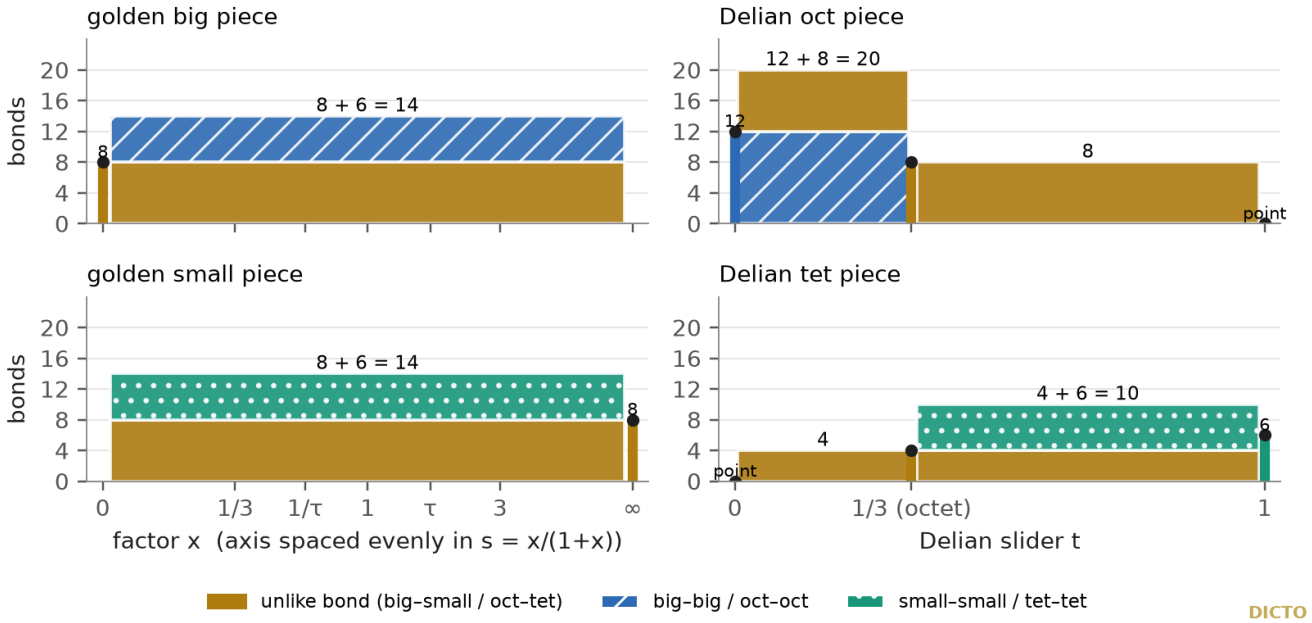


Figure 2. Bonds per piece along each loop (Corollary 1). Bars: open intervals; dots: single parameter values. The golden pieces keep 8 unlike bonds throughout and 6 like ones except at one end. The Delian octahedron piece drops from 20 bonds to 8 at the octet, and the tetrahedron piece rises from 4 to 10; at the octet both pieces have only unlike bonds.

4. Corollaries

> *The theorem tells how many bonds each piece has, where the unlike bond is strongest, and what the hidden dual lattice looks like at each moment of the loop.*

Corollary 1 (bonds per piece). Exact counts [S1, B3, C4]:

	x = 0	0 < x < ∞	x = ∞
golden big	8 big–small (octahedron)	14 = 8 big–small + 6 big–big	14 = 8 + 6 (cuboctahedron)
golden small	14 = 8 + 6 small–small (cuboctahedron)	14 = 8 big–small + 6 small–small	8 (octahedron)

	t = 0	0 < t < 1/3	t = 1/3	1/3 < t < 1	t = 1
Delian oct	12 oct–oct (rhombic dodecahedron)	20 = 12 oct–oct + 8 oct–tet	8 oct–tet	8 oct–tet	none (point)
Delian tet	none (point)	4 oct–tet	4 oct–tet	10 = 4 oct–tet + 6 tet–tet	6 tet–tet (cube)

Each golden piece's 6 squares always bond to its own kind: big–big for a big piece, small–small for a small one. What trades between the like bonds is their size, not the pieces they join; at an end, one kind's squares shrink to points.

Corollary 2 (the unlike bond's peak). The unlike bond is largest at Kelvin's cell in the golden loop, smoothly, with area $(3\sqrt{3}/16)\varphi^2$, and at the octet in the Delian loop, with a corner, with area $\sqrt{3}/4$.

Corollary 3 (the hidden dual lattice). The dual weighted-Delaunay cells are [S1, B6, B8] (volumes in units of $(L/2)^3$ for the golden loop and of the unit cube of paper #51's tetrahedron pieces for the Delian loop; exact):

Loop, range	Dual cells	Volumes
golden, 0 < x < ∞	disphenoid tetrahedra, 2 big + 2 small sites: one subdivision throughout, **no flip**	2/3
golden, x = 0 and x = ∞	square bipyramids (2 + 4 sites)	8/3
Delian, t < 1/3	octahedra (6 O) and tetrahedra (3 O + 1 Q)	4/3 and 1/12
Delian, t = 1/3	rhombic dodecahedra (6 O + 8 Q)	2
Delian, t > 1/3	cubes (8 Q) and square pyramids (1 O + 4 Q)	1 and 1/6

At t = 1/3 each rhombus of the dual rhombic dodecahedra trades its long diagonal (an oct–oct bond) for its short one (a tet–tet bond): a single flip (Figure 3). At t = 0 the dual is the octet itself; at t = 1 it is cubes.

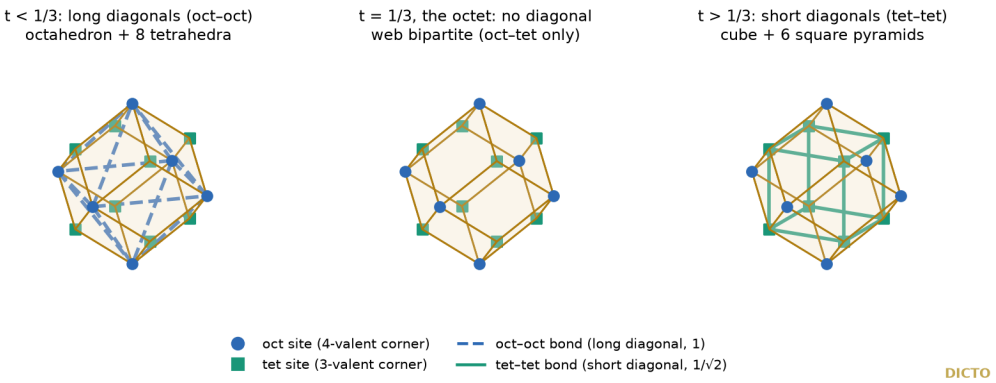


Figure 3. The hand-over at the octet, seen in the dual. One dual cell at t = 1/3 is a rhombic dodecahedron whose 4-valent corners are oct sites (blue) and whose 3-valent corners are tet sites (green). Before the octet (left) every rhombus is split by its long diagonal, an oct–oct bond of length 1, and the cell falls into an octahedron and 8 tetrahedra. At the octet (middle) no diagonal is drawn and every bond joins unlike sites. After it (right) every rhombus is split by its short diagonal, a tet–tet bond of length $1/\sqrt{2}$, and the cell falls into a cube and 6 square pyramids. Octet edge 1.

Corollary 4 (the weight line). For fixed sites, the weights that give one and the same dual subdivision form the chambers of the secondary fan [5, 6]. Each loop's weight runs along a segment. The golden segment stays inside one chamber except at its ends; the Delian segment crosses exactly one wall, at t = 1/3, and the regular subdivision on that wall is the rhombic dodecahedra.

Corollary 5 (the Delian bond). At the Delian bond, $w_Q - w_O = 3/8 - 2^{2/3}/4 \approx -0.02185$: close to, but not at, the unweighted Voronoi point t = 1/2. The ratio of its two bond areas is oct–tet / tet–tet = $2^{1/3}\sqrt{3} / (4 - 2\cdot 2^{1/3})$.

Remark (the weights in earlier work). These weights agree with paper #51's cube Remark, w = (x − 1)/(4(x + 1)) at cube edge 1, and with the octet check of #51's study; the theorem proves them for every parameter, not only at samples. The golden volumes are (4/3)(1 + u)³ − 4u³ in units of (L/2)³, with u = s for big and u = 1 − s for small, always adding to L³; the Delian volumes are those of paper #51.

5. The physical reading: balls of two sizes

> *A weight in a power diagram can be read as the square of a ball's radius: the face between two touching balls passes through their point of contact. Read this way, the two loops are two classical crystal structures, and the range where the balls can touch ends exactly at Pauling's radius-ratio limits.*

Theorem 2 (the balls). (a) The radical tessellation of a packing of balls (c_i, r_i) is the power diagram with $w_i = r_i^2 + C$, for any common constant C; a power diagram therefore fixes the radii only up to $r_i^2 \to r_i^2 + C$. (b) The golden sites are the caesium chloride structure (B2, Wyckoff 1a and 1b, lattice constant a = φ in app units); the Delian sites, shifted by (1, 0, 0)/√2, are the fluorite structure (C1: the oct sites on the face-centred cubic 4a, the tet sites on 8c, all the tetrahedral holes; a = √2). (c) If unlike neighbours touch, the radius ratio is $\rho = r_{small}/r_{big} = (x + 2)/(2x + 1) = (2 - s)/(1 + s)$ on the golden loop and $\rho = r_{tet\text{-}site}/r_{oct\text{-}site} = t/(1 - t)$ on the Delian loop; so t is exactly the share of the unlike bond length taken by the tet-site ball. (d) The touching window (unlike balls touching, like balls not overlapping) is $x \in [(\sqrt{3} - 1)/2, \sqrt{3} + 1]$, $\rho \in [\sqrt{3} - 1, (\sqrt{3} + 1)/2]$ on the golden loop, and $t \in [1 - \sqrt{(2/3)}, 1/\sqrt{3}]$, $\rho \in [\sqrt{(3/2)} - 1, (\sqrt{3} + 1)/2]$ on the Delian loop. Its ends are Pauling's radius-ratio limits [9]:

> $\sqrt{3} - 1 \approx 0.732$ for 8-coordination, and $\sqrt{(3/2)} - 1 \approx 0.225$ for 4-coordination. (e) Without the touching condition, some non-overlapping ball packing realises every $x \in [0, \infty]$, but only the $t \in [1/6, 2/3]$.

> *Why: the face between two balls is where their weighted distances agree, and that is the power bisector with weight r^2. Touching fixes the sum of the two radii, the weight fixes the difference of their squares, and the two together fix each radius.*

Proof. (a) The radical plane of two balls is the set where $|p - c_i|^2 - r_i^2 = |p - c_j|^2 - r_j^2$, that is, $2p\cdot(c_j - c_i) = |c_j|^2 - |c_i|^2 + w_i - w_j$, which depends on the weights only through their differences [S2, A1a, A1b]; for touching balls it passes through the contact point [S2, A1c] (classical: [4, 7]). (b) Exact coset comparison with the Wyckoff positions [S2, A6]. (c) Let d be the unlike nearest distance and Δ the weight difference of the Main theorem. Then $r_1 + r_2 = d$ and $r_1^2 - r_2^2 = \Delta$ give $r_1 - r_2 = \Delta/d$. Golden: $d = (\sqrt{3}/2)\varphi$ and $\Delta/d^2 = (2s - 1)/3$, so $r_{big}/d = (1 + s)/3$ and $r_{small}/d = (2 - s)/3$. Delian: $d^2 = 3/8$ and $\Delta/d^2 = 2t - 1$, so $r_{tet}/d = t$ [S2, A2, A4]. (d) Golden: the big balls touch ($2r_{big} = a = \varphi$) exactly when $(1 + s)/\sqrt{3} = 1$, at $x = \sqrt{3} + 1$, $\rho = \sqrt{3} - 1$, and the small balls touch at $x = (\sqrt{3} - 1)/2$ by the swap. Delian: the oct balls touch ($2r_O = 1$) at $t = 1 - \sqrt{(2/3)}$, $\rho = \sqrt{(3/2)} - 1$, the tet balls ($2r_Q = 1/\sqrt{2}$) at $t = 1/\sqrt{3}$, $\rho = (\sqrt{3} + 1)/2$ [S2, A2e, A2f, A4c, A4d]. (e) Put $u = r_1 + r_2$; then $r_{1,2} = (u \pm \Delta/u)/2$, both radii grow with u and ρ moves monotonically towards 1, so the feasible u form an interval, nonempty exactly when the larger radius $\sqrt{|\Delta|}$, with the other radius 0, fits. Golden: $|\Delta| \le (a/2)^2 = \varphi^2/4$ always. Delian: $-\Delta \le (1/2)^2$ exactly when $t \ge 1/6$, and $\Delta \le (1/(2\sqrt{2}))^2$ exactly when $t \le 2/3$ [S2, A3, A4f]. ∎

Corollary 6 (the stops as balls). Under the contact map: golden $x = 0$ gives $\rho = 2$ and $x = \infty$ gives $\rho = 1/2$, both outside the window, where the only packing has the other kind as points; Kelvin ($x = 1$) forces $\rho = 1$, equal balls; $x = \tau$ gives $\rho = \sqrt{5}/\varphi^2 = (3\sqrt{5} - 5)/2 \approx 0.8541$, inside. Delian: the octet $t = 1/3$ gives $\rho = 1/2$, the Delian bond $\rho = (3/2)\cdot 2^{1/3} - 1 \approx 0.8899$, the unweighted Voronoi point $t = 1/2$ gives $\rho = 1$, all inside; $t = 0$ and $t = 1$ admit no packing at all; $t = 1/6$ and $t = 2/3$ are the end packings (touching face-centred cubic balls with the tet sites as points, and touching simple-cubic balls on the tet sites with the oct sites as points) [S2, A5].

Corollary 7 (an area identity). In the Delian contact model the oct–tet triangle has area $(3\sqrt{3}/2)\cdot r_{oct}^2$ for $t \ge 1/3$ and $6\sqrt{3}\cdot r_{tet}^2$ for $t \le 1/3$ [S2, A7]. So compounds with the same oct-site radius have the same oct–tet area: BaF_2, Na_2O and K_2O, all with $r_{oct} = 1.42$ Å, share 5.239 Å².

Real compounds on the loops

Placing real ionic crystals with Shannon's effective ionic radii [8] (as transcribed by the Imperial College Database of Ionic Radii, which takes all its data from [8]; not checked against the printed page) under the contact model gives Tables 2 and 3 and Figure 4. Every row was checked exactly: the radical cell built with $w = r^2$ is the app's piece at that x or t, with the bond counts and areas of Table 1 [S2, B2]. For every compound, "in the window" is the same as "$r^+/r^- \ge 0.732$", Pauling's 8-coordination limit [S2, B3]. Areas are in app units (golden $L = \varphi$; Delian octet edge 1).

Compound	r(cation), CN	r(anion), CN	ρ	x (exact)	big–small	big–big	small–small
CsCl	Cs^+ 1.74, VIII	Cl^- 1.81, VI*	0.9613	188/167 ≈ 1.126	0.8492	0.3671	0.2897
CsBr	Cs^+ 1.74, VIII	Br^- 1.96, VI*	0.8878	109/76 ≈ 1.434	0.8412	0.4544	0.2209
CsI	Cs^+ 1.74, VIII	I^- 2.20, VI*	0.7909	133/64 ≈ 2.078	0.8155	0.5966	0.1382
TlCl	Tl^+ 1.59, VIII	Cl^- 1.81, VI*	0.8785	203/137 ≈ 1.482	0.8396	0.4666	0.2125
TlBr	Tl^+ 1.59, VIII	Br^- 1.96, VI*	0.8112	233/122 ≈ 1.910	0.8225	0.5639	0.1546

Table 2. Caesium chloride type on the golden loop: the larger ion (the anion) on the big site; every row has 8 big–small + 6 big–big bonds per big piece and 8 big–small + 6 small–small per small piece, and lies inside the touching window. Radii in Å. *Shannon gives Cl^-, Br^- and I^- only for CN VI; the CN VI value is used as a proxy for CN VIII.

Compound	oct site (CN 8)	tet site (CN 4)	ρ	t	side of octet	oct–tet	like bond	window
CaF_2	Ca^{2+} 1.12	F^- 1.31	1.1696	0.5391	> 1/3	0.2070	tet–tet 0.2610	in
SrF_2	Sr^{2+} 1.26	F^- 1.31	1.0397	0.5097	> 1/3	0.2342	tet–tet 0.2296	in
BaF_2	Ba^{2+} 1.42	F^- 1.31	0.9225	0.4799	> 1/3	0.2636	tet–tet 0.1956	in
CdF_2	Cd^{2+} 1.10	F^- 1.31	1.1909	0.5436	> 1/3	0.2030	tet–tet 0.2656	in
PbF_2 (β)	Pb^{2+} 1.29	F^- 1.31	1.0155	0.5038	> 1/3	0.2398	tet–tet 0.2231	in
UO_2	U^{4+} 1.00	O^{2-} 1.38	1.3800	0.5798	> 1/3	0.1720	tet–tet 0.3014	**out**
ThO_2	Th^{4+} 1.05	O^{2-} 1.38	1.3143	0.5679	> 1/3	0.1819	tet–tet 0.2900	in
CeO_2	Ce^{4+} 0.97	O^{2-} 1.38	1.4227	0.5872	> 1/3	0.1660	tet–tet 0.3083	**out**
ZrO_2 (cubic)	Zr^{4+} 0.84	O^{2-} 1.38	1.6429	0.6216	> 1/3	0.1395	tet–tet 0.3389	**out**
Li_2O (anti)	O^{2-} 1.42	Li^+ 0.59	0.4155	0.2935	**< 1/3**	0.3358	oct–oct 0.0794	in
Na_2O (anti)	O^{2-} 1.42	Na^+ 0.99	0.6972	0.4108	> 1/3	0.3382	tet–tet 0.1094	in
K_2O (anti)	O^{2-} 1.42	K^+ 1.37	0.9648	0.4910	> 1/3	0.2524	tet–tet 0.2086	in

Table 3. Fluorite and antifluorite on the Delian loop (radii in Å; oct site 8-coordinated, tet site 4-coordinated). In fluorite the cation sits on the oct sites; in antifluorite (Li_2O, Na_2O, K_2O) the anion does. Caution on names: the app's "oct" pieces sit on the 8-coordinated face-centred cubic sites, its "tet" pieces on the 4-coordinated tetrahedral holes.

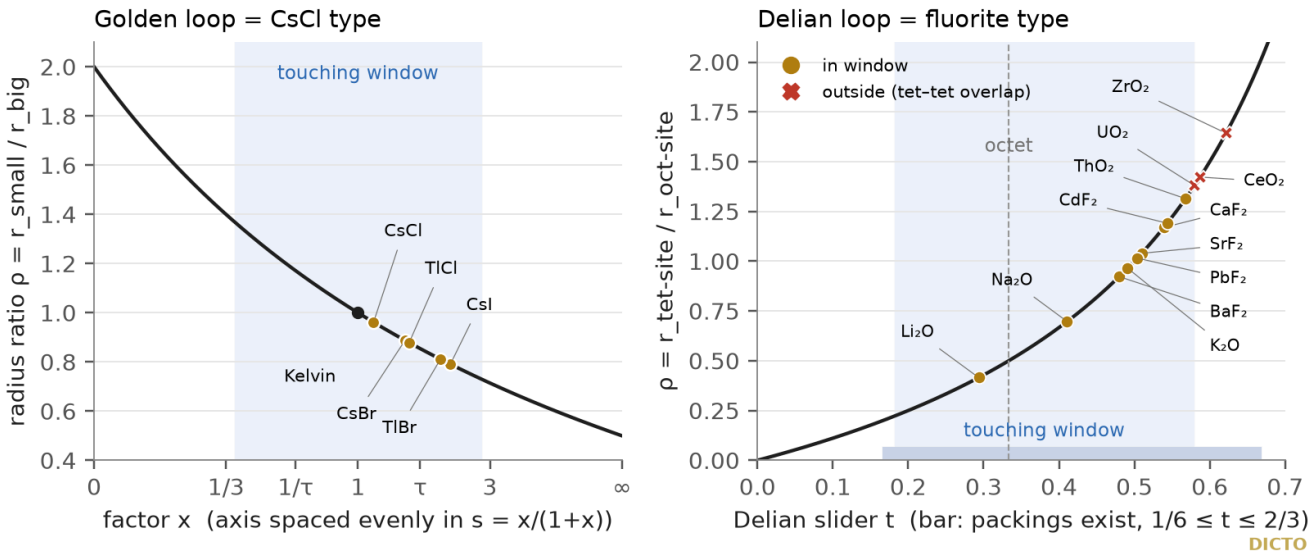


Figure 4. Real compounds on the loops (contact model; Shannon effective ionic radii [8]). Curves: the contact maps of Theorem 2(c). Shaded: the touching windows, whose ends are Pauling's limits. Left: the five caesium chloride type halides lie between Kelvin (x = 1) and the like-contact limit √3 + 1; the golden stop x = τ lies among them. Right: the fluorides and oxides; UO_2, CeO_2 and cubic ZrO_2 fall outside (red crosses). Only Li_2O lies before the octet, and that is not robust (§6).

Agreements with crystal chemistry. The sites are exactly B2 and C1. The touching windows are Pauling's limits, derived here, not quoted. The five caesium chloride type halides lie inside the window at x ≈ 1.1–2.1, on the big-ion side of Kelvin, and the golden stop x = τ ≈ 1.618 lies among them (between CsBr/TlCl and TlBr/CsI). Each golden piece's 14 faces (8 unlike + 6 like) match the two body-centred cubic neighbour shells, 8 at (√3/2)a and 6 at a. Cubic ZrO_2 lies far outside the window ($r^+/r^- = 0.61$), which fits the general knowledge that cubic zirconia is not stable at room temperature unless doped.

6. Limits of the hard-ball reading

To be updated from the scale-anomaly study (2026-10-25, running). The known misfits, as found in [S2]:

> *The balls are a reading of the geometry, not a model of the crystals. Where real crystals and hard balls disagree, the geometry stands and the reading stops.*

UO_2 ($r^+/r^- = 0.725$) and CeO_2 (0.703) are stable fluorites, yet with effective ionic radii they fall outside the touching window: both the hard-ball contact model and Pauling's rule miss here. With crystal radii (CR = IR ± 0.14) both fall inside, so the verdict depends on the radius convention; under crystal radii UO_2, CeO_2 and ZrO_2 move into the window (t = 0.521, 0.528 and 0.559). BaF_2 has $r^+/r^- = 1.08 > 1$, where Pauling's rule would prefer a higher coordination; the model, which tests only overlap, has no objection. Na_2O (0.697) and K_2O (0.965) lie far above the 0.414 upper limit for 4-coordination, yet both are antifluorites; the model allows them, as it has no upper "rattling" limit. Li_2O moves under crystal radii from t = 0.2935 to t = 0.363, past the octet, so its side of the octet stop is not a conclusion; for every other compound the side is the same under both conventions. CsCl moves from x = 1.126 to 1.432 under crystal radii, the other halides towards x = 1. The radical faces are not the distance shells: in fluorite with t > 1/3 the cation has 12 cation neighbours at a/√2 but no cation–cation face; the anions' cubes separate the cations. β-PbF_2 is placed, but ambient PbF_2 is the orthorhombic α (cotunnite type) form.

Limits of the model: real ions are not hard balls; a power diagram fixes only $r_1^2 - r_2^2$, and the radius ratio needs the contact assumption; the real lattice constant is set by the crystal, not by $r_1 + r_2$, and a placement from measured lattice constants was not made; the halide radii are CN VI proxies; the radii come from a transcription of Shannon's table. Left out: NH_4Cl (no NH_4^+ entry in the transcription), the metallic B2 alloys CuZn, NiAl, FeAl, CoAl (metallic radii unverified), Rb_2O (no Rb^+ CN IV entry).

7. Conclusion

> *DICTO's web of bonds is a lattice that changes with the loop, as he proposed: the classical dual of a weighted Voronoi diagram. In the golden loop it changes only in size; in the Delian loop it changes its shape once, at the octet.*

Both DICTONIC loops are one fixed crowd of points with one weight slid between two kinds, and the web of bonds is the dual weighted-Delaunay web. Its rhythm differs between the two loops. In the golden loop the dual never flips: the like bonds trade size in the ratio x^2, equal at Kelvin's cell, and the whole loop is symmetric under $x \leftrightarrow 1/x$. In the Delian loop the dual flips once, at the octet, where every rhombus of a hidden rhombic dodecahedron exchanges its long diagonal for its short one, and the octet is the one member whose web joins only unlike pieces. Read as balls, the loops are the caesium chloride and fluorite structures, and Pauling's classical limits mark where the balls can touch. Open: the scale-anomaly study (§6); the keyed bond's web; a web search before any claim of novelty.

8. What is known, and what is new

> *Power diagrams, their duality with weighted Delaunay subdivisions, radical tessellations of ball packings, the two crystal structures and Pauling's rule are all classical. What we did not find is their application to these two loops.*

Known. Power (Laguerre, weighted Voronoi) diagrams and their duality with regular triangulations [3, 4]; regular subdivisions change only at the walls of the secondary fan [5, 6]; radical tessellations of packings of balls of two or more sizes [7], and Voronoi–Dirichlet and radical polyhedra in crystal chemistry and coordination [18, 19, 21]; two unequal truncated octahedra on a body-centred lattice as Laguerre cells (general; noted in paper #48 [1]); the octet, rhombic dodecahedron and cube connections [11]; the T(x) orbit [10]; ionic radii [8] and the radius-ratio rule [9]; caesium chloride type and other binary crystals of colloids and nanoparticles whose phase is set by the size ratio [12–17]. Paper #51's cube Remark [2] and its octet check already gave the two weights at samples. Voronoi's theory of Delaunay subdivisions of lattices (Erdahl [22]) is related, but treats lattices, not two kinds of weighted sites.

Not found, in the sources listed in Appendix A: the closed-form bond areas of both loops; the contrast between the golden loop (no flip) and the Delian loop (one flip, at the octet); the like-bond ratio x^2 and the diagonal sum φ; the sequence of dual cells along each loop; the octet as the only member with a bipartite web; the contact maps $\rho = (x + 2)/(2x + 1)$ and $\rho = t/(1 - t)$; the touching windows bounded exactly by Pauling's limits; the Delian packing range $1/6 \le t \le 2/3$. These are elementary consequences of standard theory and may well be known in some form; "not found" is limited to those sources, and a web search is still needed before any claim of novelty.

9. In the app

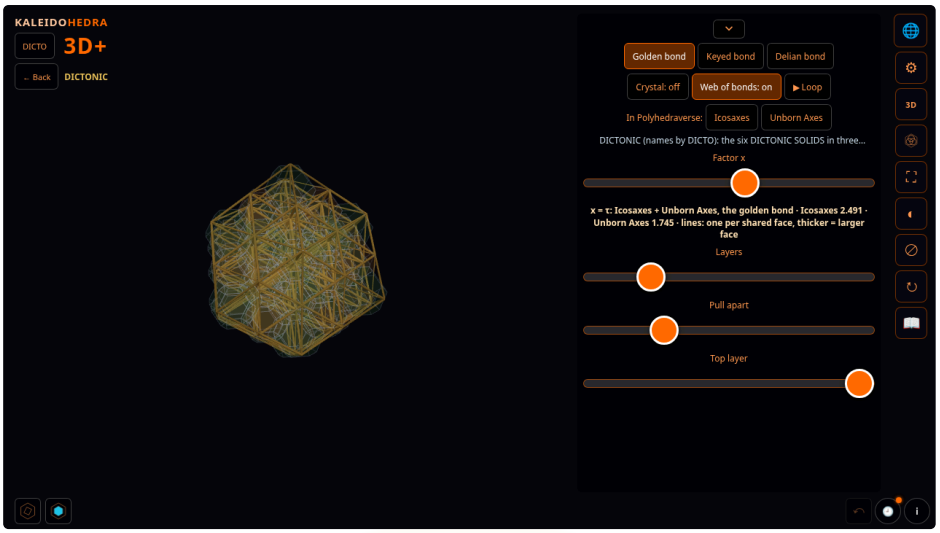


Figure 5. App screenshot (Kaleidohedra, DICTONIC world, golden bond at x = τ, "Web of bonds: on"; https://dictospheres.com): one line per shared face, thicker for a larger face. The ▶ Loop button runs the factor through the loop, and the web changes with it.

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[S1] DICTO, exact study of the web of bonds (studies/2026-10-25-bond-relay, commit 7b0fd8b): scripts/symbolic.py → data/checks/symbolic-checks.txt (part A, whole regimes, 55 checks), scripts/samples.py → samples-checks.txt (part B, 10 golden and 11 Delian parameters, 211 checks), scripts/app_dump.mjs + check_app.py → app-checks.txt (part C, the app's own pieces and web, 105 checks).
[S2] DICTO, physical anchors study (studies/2026-10-25-physical-anchors, commit 7b0fd8b): scripts/symbolic.py → symbolic-checks.txt (part A, 32 checks), scripts/compounds.py → compounds-checks.txt and data/raw/compounds.json (part B, 29 checks), scripts/crossref_check.py → data/raw/crossref.json.

Appendix A. Sources checked for prior work

DICTO's reference library (kaleidohedra-references): nine ask-library queries (power diagram / Laguerre / weighted Voronoi; contact graph of a space-filling; the Koca truncated-octahedron family on a body-centred lattice; the octet / rhombic dodecahedron / cube family; regular triangulations, flips and the secondary polytope; caesium chloride type binary colloidal crystals with tunable size ratio; radical tessellation and Laguerre–Voronoi polyhedra of sphere packings in crystals; ionic radii, radius ratio, coordination and Pauling's rules; the fluorite structure and the tetrahedral holes of the face-centred cubic lattice), and its index, reading list and README searched by keyword. Nothing was found there on power diagrams, Laguerre or radical tessellations, contact graphs, secondary polytopes, ionic radii or radius-ratio rules; the nearest entries were Koca et al. 2015 [10], Erdahl 1999 [22] and Garber 2019 (Voronoi's theory for lattices), and Cadotte, Dshemuchadse, Damasceno, Newman and Glotzer 2016 (binary hard tetrahedra and octahedra, not balls). Web, 25 October 2026: one search for radical tessellation in ionic crystals, which found no paper linking power diagrams with ionic radii, caesium chloride or fluorite. References [3–5, 7–9, 12–19, 21, 22]: every DOI checked on Crossref (title, journal, year, volume and pages); none of these papers was read in full for this paper. References [6] and [20] are cited from general knowledge, [11] as in paper #51. Not checked: a full web search on the bond-area closed forms and the dual-cell sequence.

Appendix B. Checks

Study	Part	Checks	PASS	FAIL
web of bonds [S1]	A symbolic, whole regimes (A1–A7)	piece = power cell, areas, volumes, face to face, monotonicity, distance bound	55	0
	B exact samples (B1–B8)	cells from scratch at 10 golden and 11 Delian parameters, rational and irrational	211	0
	C app cross-check (C1–C4)	frame, every app bond area, counts, float findings	105	0
physical anchors [S2]	A symbolic (A1–A7)	weight = r^2, maps, windows, feasibility, stops, B2/C1, area identity	32	0
	B exact cells (B0–B3)	Kelvin and rhombic dodecahedron sanity, 9 loop stops, 17 compounds, Pauling = window	29	0
total			**432**	**0**

Exact arithmetic: Python fractions over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{6})$, $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\sqrt[3]{6})$, and sympy for everything that depends on the parameter; a number is zero only if all its field coefficients are; floats only choose candidate plane

triples and serve as the app cross-check. Golden samples x = 0, 1/3, 1/2, 1, 2, 3, τ, 1/τ, √2, ∞; Delian samples t = 0, 1/6, 1/4, 1/3, 1/2, 2/3, 1 and the stops $(2\sqrt{6}-3)/15$, $1-\sqrt[3]{6}/3$, $1-(2/3)\cdot 2^{-1/3}$, $5/3-(2/3)\sqrt{2}$.